

Mindful Walk: cue card

1 Original concept: Tooth brushing

- Bathroom; two minutes, twice a day
- This is all the robot said: instructions
- No room for theory of mind
- Needs a steady grip; if the hands fail, the concept fails

2 Think of your last vacation.

- Ask the room; wait twenty seconds
- Two or three people share
- Then, out loud: "Hands up if you were moving: walking, hiking, swimming, cycling"
- Count the hands

3 Moving helps memory

- Link back to the hands
- Walking AR tour against a seated VR tour of the same museum
- Large effect: $d = 1.31$, 28 adults
- Tested straight after and 48 hours later

4 More to talk about

- Walks bring residents together: partner, volunteer, family
- Every walk differs: season, weather, stories
- Less repetition in the final presentation

5 What a walk can serve

- Six Human Values from our Foundation
- Company: a partner with a shared past
- Identity: the old round, the old songs
- Autonomy: whether, where, how long
- Well-being, dignity, safety

6 Mindful Walk

- Walk: daily, outdoors, with company
- Notice: feet, air, birdsong
- Remember: the life story along the route
- Rest and listen: rain days, dusk
- Walking comes first

7 SA-01 · Accompanied walk

- Today's fix: closed garden loop, coded exits, family and volunteers
- The companion is the resident's anchor
- Addressed by: present-moment cues, gate alert

8 SA-02 · Fitting walks between care tasks

- Walks compete with care tasks for the same staff time
- Addressed by: walks with a partner and the guide; staff in reach

9 SA-03 · Bad-weather indoor exercise

- The exercise stays; daylight, sights and company go
- Addressed by: seated practice at the window

10 SA-04 · Walking with a fellow resident

- The match decides whether the walk works
- Addressed by: pairing on a shared past; memory prompts

11 SA-05 · Relaxation session

- Recordings are made for general audiences
- Addressed by: concrete cues, paced to the resident

12 SA-06 · Winding down to rest

- Restlessness peaks when staff time is lowest
- Addressed by: lying practice with the docked guide

13 SA-07 · Listening to radio, news or audiobooks

- Preferences known to family, not to whoever switches on
- Addressed by: two-option offers from the life story

14 Stakeholders

- Eight stakeholders; ST-08 is new for this concept

15 Problem scenarios

- Three on the walk, three at rest

16 Human values

- Five carried over from morning care; Safety, Social connectedness and Identity are new

17 Value tensions

- Right column: how the design resolves each tension

18 Human factors concepts

- HFC-01 is the vacation question; the recall task tests it

19 Measures

- Validated instruments where they exist: OERS, OME, CMAI

20 Evaluation methods

- Detailed in the Evaluation section

21 Technology options

- Five selected, four rejected; the rejections are part of the rationale

22 What we keep from morning care

- New activity, same principles
- Graduated prompting became cue, then silence
- The care-worker handover became the alert
- Three-item record then, four items now
- Also kept: staged tests, the within-resident comparison, adverse claims

23 Jan

- Our persona
- Postal worker: thirty years, on foot and by bike
- Moderate dementia: today is gone, the round is vivid
- Refused a GPS watch: "I'm not a parcel"
- Kees, two doors down, sorted the post at the same depot
- On the walk: a memory prompt at the rose bed; they talk for ten minutes

24 Voice in the pocket, person in reach

- Pepper indoors only: no grass, no gravel
- Pocket guide outside, same voice
- Beacons at pond, roses, bench and gate
- No GPS, no route stored; microphone off outdoors
- A person always walks along

25 How the voice speaks

- Three rules
- Adult words, normal pitch
- No praise, no endearments

26 Rest, seated or lying

- Rain: Pepper at the window
- Breathing light on Pepper's shoulders
- Dusk: guide docked at the bedside
- Concrete cues: nothing to remember

27 Listening that fits

- Two options from the life story
- News in the morning, calm in the evening
- News always on request, never hidden
- Short stories; a recap for long books

28 What could go wrong

- Memories can hurt: "I must go home"
- The voice may sound childish
- The robot may feel like being watched
- Calm evenings may feel like censorship
- Each one tested; if true, the design changes

29 Design scenarios

- Same resident, Jan, in all three

30 Personas and robot profiles

- Four people, two devices, one voice

31 Objective stories

- Each ties a Function to a value, in a stakeholder's words

32 Objectives

- Four Musts; walking comes first

33 Use cases and functions

- Three use cases share six functions

34 Claims

- Fourteen claims, four adverse
- Success criteria in the Evaluation section

35 Design patterns

- One team pattern, three interaction patterns

36 Evaluation

- Verification: do the Functions work as specified (Premises)
- Validation: do they have the intended effect (Claims)
- Four phases, thirteen weeks

37 Evaluation plan

- Phase 1: technical checks, no residents
- Phase 2: human-operated prototype, tune the voice
- Phase 3: ABAB validation of the Claims
- Phase 4: interviews with staff and family
- Ethics approval before any resident takes part

38 Phase 1: technical verification

- Staged events against a timestamped reference log
- Premises PR1-PR6
- Numbers are proposals; care workers set the final ones

39 Phase 2: human-operated prototype test

- Formative: before anything is automated
- Each factor varied across short walks and sessions
- The wording residents respond to best goes into the script

40 Phase 3: summative validation

- Single-case ABAB: baseline, intervention, baseline, intervention
- 8-12 residents, each their own control
- Walk data every walk; observer on sample walks
- Agitation rated by staff at the end of each phase
- Recall task and interview in intervention phases only

41 Recall task: ambulatory versus seated encoding

- The vacation question, tested on residents (CL3)
- Same target event: a bell, walking or seated
- One hour later: free recall, then a choice of three pictures
- Stop at any sign of discomfort

42 Success criteria

- Each criterion is set before the study starts
- Staffing counts only if human company stays the same
- A null result on recall weakens the walking-first argument; we report it either way